MATLAB TECHNICAL REPORT

Mathematical Hydropower Plant Model and Performance Analysis

Çalışma / Project	Digital Twin Study Referencing Kavşak Bendi HEPP
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Kapsam / Scope	Educational and engineering portfolio simulation

Abstract

This report presents a mathematical hydropower plant model and a set of performance analyses developed in MATLAB with Kavşak Bendi Hydropower Plant (HEPP) used as a reference. The model examines hydraulic energy conversion, the effects of flow rate and net head, reservoir water balance, unit operating schedules, turbine-generator efficiencies, transformer and grid-interface calculations, illustrative protection scenarios, and alternative operating conditions.

The work is intended for educational and engineering-portfolio purposes. Some plant reference ratings were supplied as project inputs, while several operating parameters—including flow, head, efficiency, reservoir assumptions, and protection thresholds—are modelling assumptions. The results must not be interpreted as verified plant operating data or SCADA measurements.

1. Objective and scope

The objective is to represent the main energy-conversion chain of a hydropower plant mathematically and investigate how operating variables affect electrical power and daily energy production. This is a simplified engineering model based on reference information, not a complete replica of the plant's hydraulic and electrical design.

- Hydraulic and electrical power calculations
- Flow-rate and net-head sensitivity analyses
- 24-hour generation and energy calculations
- Reservoir volume and water-level balance
- Turbine availability schedules and efficiency curve
- Generator, transformer, and grid-interface calculations
- Illustrative protection and fault scenarios
- Comparison of operating points and degradation cases

2. Reference plant and model parameters

The reference installed ratings used in this project are three main Francis units rated at 61.95 MW each and one ecological-flow unit rated at 5.43 MW, giving a total reference installed capacity of 191.28 MW. These values were used as project inputs and were not independently verified in this report.

Parameter	Value	Description

Number of main units	3	Francis turbines
Main unit rating	61.95 MW/unit	Reference input
Ecological unit rating	5.43 MW	Reference input
Total installed capacity	191.28 MW	3×61.95 + 5.43
Gross head	100 m	Model assumption
Hydraulic losses	5 m	Model assumption
Net head	95 m	100 − 5
Turbine efficiency	0.90	Model assumption
Generator efficiency	0.96	Model assumption
Overall efficiency	0.864	0.90×0.96
Reference total flow	237.55 m^3/s	Model input; not verified as design flow

3. Mathematical method

Net head is calculated by subtracting hydraulic losses from gross head:

$H_n = H_\beta - h_k = 100 - 5 = 95$ m

The power-conversion chain is represented by:

$P_h = \rho \cdot g \cdot Q \cdot H_n$

$P_m = P_h \cdot \eta_t$

$P_e = P_m \cdot \eta g = \rho \cdot g \cdot Q \cdot H_n \cdot \eta_t \cdot \eta g$

Here, ρ is water density (1000 kg/m^3), g is gravitational acceleration (9.81 m/s^2), Q is flow rate (m^3/s), H_n is net head (m), η_t is turbine efficiency, and ηg is generator efficiency. The factor 10^{-6} converts power from watts to megawatts.

4. Analyses completed

Analysis	Implementation / main outcome
Power and flow distribution	Calculated unit-level power and flow distribution and generated flow-power plots.
Net-head sensitivity	Investigated net head over a 70–110 m range, with output limited by installed capacity.
24-hour operation	Created hourly flow and power profiles and calculated daily energy and average power.
Reservoir balance	Modelled volume and water-level changes using $V(t+\Delta t)=V(t)+(Qin-Qout)\Delta t$.
Unit operating schedule	Evaluated example schedules in which a main turbine was unavailable during selected periods.
Francis efficiency curve	Used an illustrative flow-dependent efficiency curve with interp1 and calculated a power curve.
Generator	Calculated active/reactive power, apparent power, power factor, and current.
Transformer	Performed an illustrative loading and loss calculation for a single-unit transformer reference: 13.8/154 kV, 80/100 MVA ONAN/ONAF.
Grid interface	Evaluated active/reactive power, apparent power, power factor, and high-voltage-side current.

Protection scenarios	Tested illustrative overcurrent, undervoltage, underfrequency, and transformer-overload cases.
Operating points	Compared 50%, 75%, and 100% load conditions.
Integrated model	Evaluated the effects of flow, net head, and efficiency changes on hydraulic, mechanical, and electrical power.

5. Key numerical results

Indicator	Result	Note
Reference hydraulic power	≈221.39 MW	$\rho g Q H_n$ calculation
Reference mechanical power	≈199.25 MW	At 90% turbine efficiency
Reference electrical power	≈191.28 MW	At 96% generator efficiency and installed-capacity limit
Daily energy (example profile)	≈3,847.3 MWh	Depends on hourly operating schedule
Average power (example profile)	≈160.31 MW	24-hour average
Daily capacity factor	≈83.81%	Example profile
Daily energy at constant full load	4,590.72 MWh	191.28 MW × 24 h; a different scenario

The values 3,847.3 MWh and 4,590.72 MWh represent different operating assumptions. The former comes from an example schedule with reduced output or unavailable units during some hours; the latter assumes continuous operation at full installed capacity for 24 hours.

6. Assumptions and limitations

- A significant portion of flow, head, and efficiency inputs are illustrative model parameters, not verified design or operating data.
- Electrical output is capped at the installed-capacity value in the normal operating calculation. This is a simplified capacity constraint.
- The reservoir model uses a simplified volume balance; actual elevation-area-storage curves and hydraulic transients are not represented.
- The transformer calculation represents an illustrative single-unit connection. The actual plant single-line diagram and transformer arrangement require separate verification.
- Protection thresholds are educational examples, not actual relay settings or operating procedures.
- The model does not include detailed load-flow, short-circuit, transient-stability, or real-time SCADA integration.

7. Conclusion and future work

The MATLAB work provides a starting mathematical model for examining the hydropower energy-conversion chain and key operating variables within one framework. The analyses demonstrate the influence of flow, net head, efficiency, and unit availability on generation. The next steps are to extend the Simulink model with dynamic signal behaviour and then continue in Google Colab with data analysis, synthetic sensor data, performance monitoring, and visualization.

Data and source note

This report is based on reference plant information supplied by the user and modelling assumptions adopted during the project. External source verification and access to actual plant operating data were not part of this report.